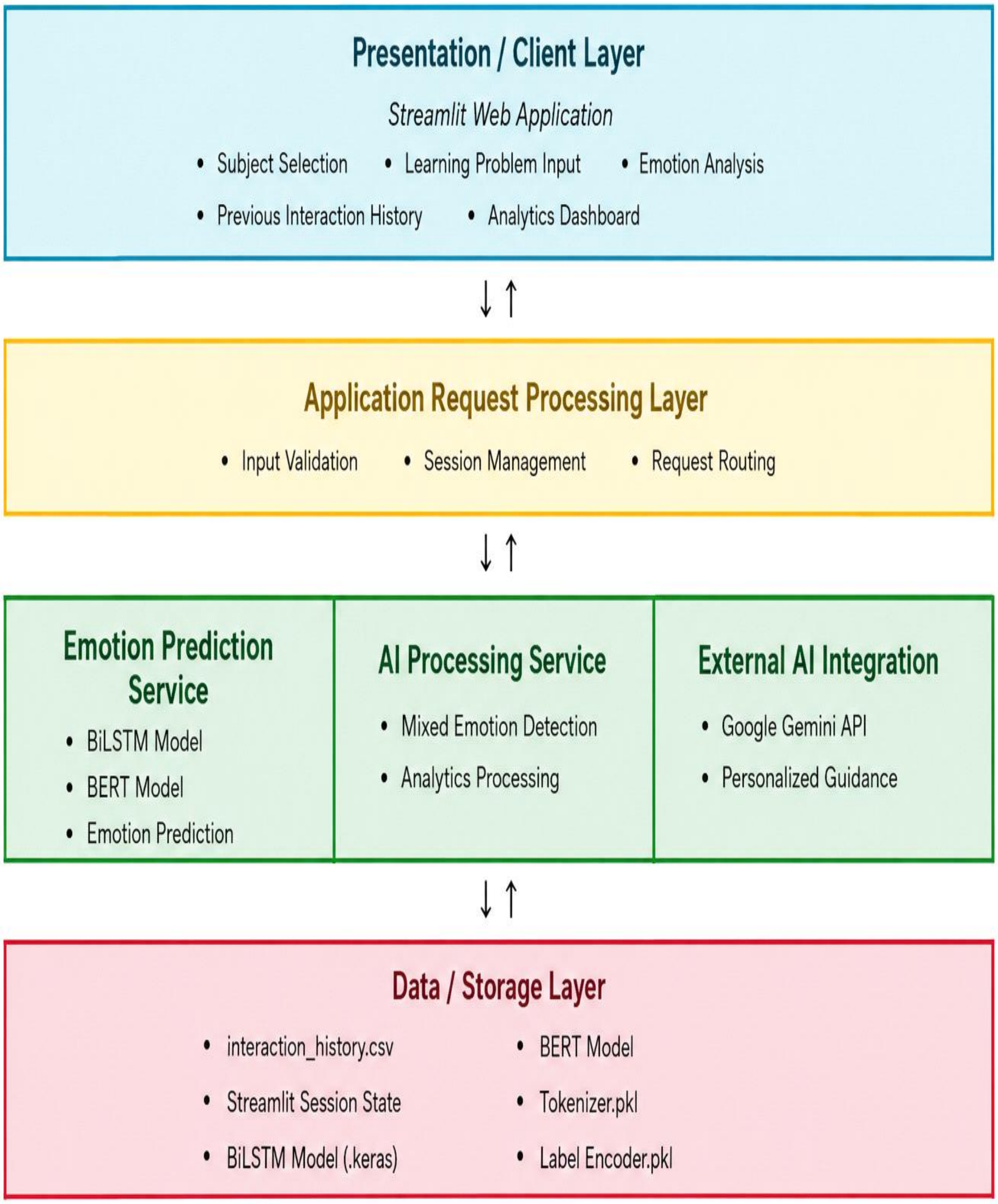
Solution Architecture

Date	15 March 2026
Team ID	xxxxxx
Project Name	AI Learning Support Assistant
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the user interface where students	Streamlit, Plotly

	select a subject, enter their learning problem, view emotion analysis, AI guidance, previous interactions, and analytics dashboard.	
Application Request Processing	Validates user input, manages session data, coordinates prediction flow, and sends requests to AI models and the Gemini API.	Python, Streamlit Session State
Core Logic Service	Performs text preprocessing, BiLSTM emotion prediction, BERT emotion prediction, mixed emotion detection, prediction schema generation, interaction logging, and analytics processing.	Python, TensorFlow, Hugging Face Transformers, Pandas, NumPy, Scikit-learn
External API Integration	Generates personalized learning recommendations based on the detected emotion and learning problem.	Google Gemini API
Data / Storage Layer	Stores interaction history, trained models, tokenizer, label encoder, and session information used by the application.	CSV (`interaction_history.csv`), Streamlit Session State, TensorFlow Models, Pickle

Architecture Workflow

- The **student** opens the Streamlit application.
- The student selects a **subject** and enters a **learning problem**.
- The application validates the input.
- The text is processed by the **BiLSTM** and **BERT** emotion prediction models.
- The **Mixed Emotion Detector** identifies the primary and secondary emotions.
- The detected emotion and learning problem are sent to the **Google Gemini API** to generate personalized learning guidance.
- The interaction is saved to **`interaction_history.csv`**, and the current session is managed using **Streamlit Session State**.